

MAT 112 2.0 Differential Equations

Tutorial 06

1. Find the Wronskian of

1. $\{\sin 3x, \cos 3x\}$

2. $\{3x, 5x\}$

3. $\{e^x, e^{-x}\}$

4. $\{1, x\}$

5. $\{e^x, xe^x\}$

2. Using the Wronskian find linearly independent solutions of the following equations.

1. $\frac{d^2 y}{dx^2} + 6\frac{dy}{dx} + 9y = 0.$

2. $\frac{d^2 y}{dx^2} + 3\frac{dy}{dx} = 0$

3. $\frac{d^2 y}{dx^2} + 4y = 0$

3. Show that any solution of $\frac{d^2 y}{dx^2} + y = 0$ is a linear combination of $\cos x + \sin x$ and $\cos x - \sin x$.